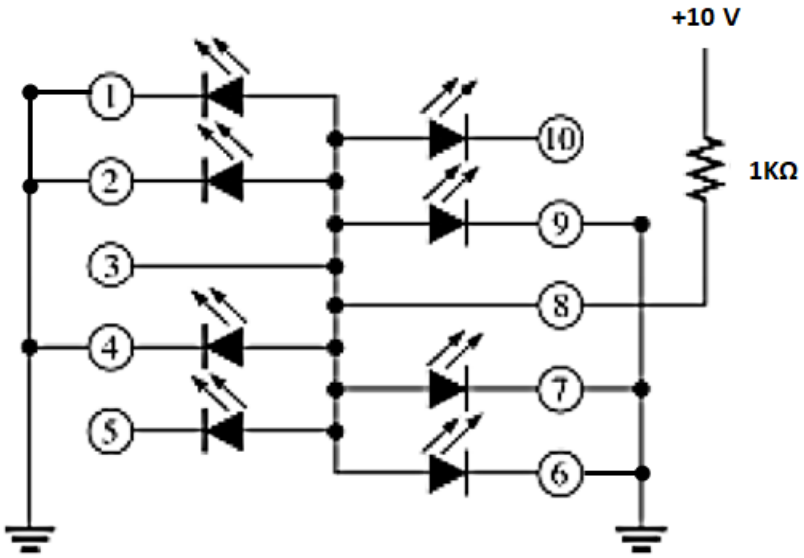
Model answer of sheet 5

1

$$R = \frac{V_s - V_f}{I_f} = \frac{10 - 2.1}{7.9mA} = 1K\Omega$$

- All anodes to Vs
- Connect cathode of (A, B, C, D, E, F) to ground with pins (7, 6, 4, 2, 1, 9)



2

$V_F \cong 2.5$ V for $I_F = 30$ mA

Maximum LEDs/branch $= \frac{24\text{ V}}{2.5\text{ V}} \cong 9.6$

Select 5 LEDs/branch:

Number of branches $= \frac{50}{5} = 10$

$R_{LIMIT} = \frac{24\text{ V} - 5(2.5\text{ V})}{30\text{ mA}} = 383\ \Omega$

